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Hybrid Predictive Framework for Expiry Date Estimation of Live-Attenuated Vaccines Using Stability Testing and Machine Learning

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***Abstract*—Transport sustainability is a major issue for most pharmaceutical agents, especially live-attenuated vaccines, which are highly sensitive to the environmental factors like temperature, humidity, light, and oxygen which is why there is need of of serious steps to be taken in the supply chain. In pharmaceutical supply chains, the most important part is to enhance the predictions related to stability for better storage and control over the inventory and minimized waste. Even though the supply chain helps us in tracking the Vaccine, the condition of it is still unknown. The stability of the vaccine is tested via these stability tests. Any deviation from the optimal order at any stage in the supply chain can compromise stability and make vaccines ineffective or unsafe. This study takes care of this by performing stability testing predicting the drug expiry according to the environmental factors. The predictive framework is a combination of machine learning regression models and the Arrhenius Equation to accommodate environmental variables affecting degradation. Ambient temperature and humidity were identified as the most significant environmental factors affecting degradation rates. The hybrid model comprising both ElasticNet and LSTM showed better prediction performance with a mean absolute error (MAE) cut off under 4.5% as compared to other conventional methods. For example, Gradient Boosting Regressor (GBR) attained an R^2 score of 0.85, the conventional methods scoring between 0.60 and 0.75. The model also accounted for a 12% decrease in the errors of expiration prediction when exposed to exposing temperature fluctuations during transportation. This is essentially intended to be produced in terms of standard recommendations for stability testing, which are crucial to the life cycle for the safe and effective use of medications.**

***Index Terms*—Stability Testing, Arrhenius Equation, Machine Learning, Live Attenuated Vaccines**

I. Introduction

Live attenuated vaccines used a weakened form of a virus so that an immune response can take place without causing disease. Therefore, their chemical compositions (by the viral antigens, stabilizers, and preservatives) are vulnerable to degradation under less-than-optimum environment conditions. [1] The stability of these vaccines is imperative for their efficacy and safety to the patient upon administration. The environmental factors assume greater urgency vis-a`-vis Live Attenuated Vaccines once they outlast the supply chain's existence. The pharmaceutical supply chain contains several stages, ranging from manufacture sites to warehouses, transport networks, hospitals, home pharmacies, and patients, among others. Therefore, it becomes critical to keep environmental conditions by temperature, humidity, exposure to light, and oxygen at each stage to keep the integrity of the drugs intact. Failure to maintain such variables optimally leads to chemical degradation of drugs thereby compromising their potency and therapeutic effectiveness. Such degradation therefore poses health hazards to the patients and financial loss to both companies and stakeholders along the supply chain. Live attenuated vaccines are vaccines that promote the immune response using microbial components that have been modified to become less infectious or virulent. Temperatures that go above or below the recommended conditions affect the viability and potency charts of the vaccine. Again, it is quite tough to develop a cold chain in areas where easy access to

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reliable refrigeration is not available. [2] In a study conducted on the stability of Esomeprazole, researchers predicted the impact of environmental stresses like temperature and oxygen exposure on drug stability via a deep learning model. Taking such conditions as pH into consideration and using artificial neural networks, researchers obtained critical ranges which would be a safeguard for ensuring the quality of drugs over a shelf life of 36 months [3]. Indeed, cold chain modern logistics marketing has made strides in ensuring that live-attenuated vaccines are intended for use without deteriorating along the supply chain. The effect of distribution on the typical stability test scenario-real time and accelerated stability tests, cannot be replicated adequately. Along with the installation of real-time environmental monitoring, it was further noticed that there was a gap in the prediction of degradation. It is, therefore, necessary to pursue new ways, which incorporate predictive modeling along with stability data. [4] [5]

In the present study, we demonstrated a new approach combining the traditional Arrhenius mathematical model and machine-learning regression techniques to infer stability and expiry dates of live-attenuated vaccines in varying environmental conditions. [6] The Arrhenius equation describes the temperature dependence of reaction rates and thus forms the basis on which chemical degradation processes can be modeled. Integration of this with machine learning regression models should result in automating and improving the accuracy of stability predictions for better decision-making in the distribution and storage of vaccines. [7]

This study is concerned with the possibility of developing improved public health outcomes and efficaciousness in operations by solving the challenges presented in large-scale applications with live attenuated vaccine supply into the chain.

A. Problem Statement

The research problem here addresses the fact that the environmental changes are so unpredictable during the transport and storage of live-attenuated vaccines that actually determining their expiration date is quite tricky. Stability testing methods from the past involved real time and accelerated stability tests, mostly under controlled laboratory conditions, to estimate how long a vaccine would remain stable. However, all these methods do not find space for the environment in terms of the fluctuations that suddenly occur as far as geographies across the global supply chain. Thus, research was an attempt to realize all the necessary parameters that would influence the stability of drugs, such as temperature, humidity, exposure to light, and the presence of oxygen. The combination of the Arrhenius equation and regression-based machine-learning models helps establish a robust predictive framework that can accurately estimate expiry dates of drugs in different environmental conditions.

The primary objectives of this study include:

- **Constructing a predictive model by means of environmental factors such as temperature, humidity, light, and oxygen in order to forecast drug expiry.**
- **The use of stability testing methods to investigate the degradation patterns of a drug. For example, regression-based machine-learning models and the Arrhenius equation.**
- **Maintain and ensure safety to the Public.**

II. Literature Review

This literature review provides insights derived from an extensive reference to research papers closely aligned with our problem statement. The review covers key themes, including advanced algorithms for predictive modeling, stability testing techniques, and the critical importance of live attenuated vaccines. The studies explored highlight innovative approaches for pharmaceutical stability prediction, supply chain optimization, and the use of emerging technologies like blockchain and machine learning to enhance vaccine safety, distribution efficiency, and public health outcomes. These findings form a solid foundation for our research direction and problem-solving approach.

Cle´net et al. [8] employs two-step kinetic modelling to predict vaccine stability during storage and temperature excursions. It validates degradation predictions for freeze-dried vaccines under varying conditions, ensuring reliable distribution. The study emphasizes advanced predictive models for optimizing vaccine transportation and storage, addressing challenges in cold-chain management. The study [9] applies kinetic modeling to determine the stability of mRNA vaccine by using real time temperature data and monitoring trends. It addresses the cold chain challenges, especially in the low income countries by establishing The products' shelf life along with critical quality attributes for vaccines. The approach enhances the management of the vaccine stock, minimizes wastage and guarantees safety of the vaccines regardless of the distribution condition.

Tembhare et al. [10] highlights emphasis various on stability the of testing real pH methods time and that and temperature are accelerated on used stability the in testing. stability order It of to also drugs establish explains and the the the shelf role various analytical tools like LC-MS/MS and HPLC for stability evaluation. The study [11] also describes nonparametric bootstrap approaches alongside one-stage non-linear Arrhenius models for determining pharmaceutical shelf-life. It also shows that the bootstrap methods provide better coverage and accuracy of the estimated shelf-life especially when applied under accelerated stability conditions than the conventional methods. In [12] research study, AKM is assessed for its efficacy in assessing the long term stability of biotherapeutics and vaccines. The study further reveals that AKM can effectively estimate stability for up to three years even with variation in temperature thus positioning it as a viable tool for stability forecasting.

In the study [13], several machine learning algorithms such as Linear Regression, Random Forest, Support Vector Regression, and XGBoost are evaluated for the task of pharmaceutical drug sales prediction. The results show that XGBoost

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outperforms other algorithms in dealing with big data and providing precise estimation. This research also stresses the need for data-based strategies for improving drug stock management and reducing waste. Habiyaremye et al. [14] research propounds a predictive machine learning model that can be used to manage the storage of temperature sensitive products for instance vaccines in multi chambered refrigerators. It uses multiple linear regression to determine the remaining time before certain temperatures are reached and exceeded thus unsafe for storage. The model has the potential of greatly improving the reliability of the vaccine storage systems in pharmacies. In [15], the authors present new statistical models for estimating the shelf life of pharmaceuticals while addressing the non-linear stability trends. Thus, it applies the random coefficient mixed regression models and extends the tolerance interval approach to account for the between-and within-batch variations. The study also emphasises the ability of nonlinear models in the assessment of pharmaceutical stability as it presents robust options to the linear models.

Liu et al. [16] study is an attempt to develop a blockchain based vaccine supply chain model to analyse competi- tion between manufacturers and their operational decisions. Blockchain improves the efficiency of the cold chain, makes the process of tracking and monitoring the vaccines easier and makes the vaccines safer. The paper suggests revenue-sharing and two-part tariff contracts as the coordination mechanisms to improve the supply chain performance and increase the stakeholders' profits while maintaining the quality of the vaccine and the trust of the public. Haji et al. [17] review highlights challenges in food and drug supply chains, focusing on quality, safety, and counterfeiting. It explores the potential of modern tools like blockchain and IoT to create resilient, integrated supply chains. Emphasizing cross-verification and resource management, the paper identifies strategies to ensure sustainability, public health, and the reliability of food and pharmaceutical systems. In [18] aims at studying the role of AI and ML in the transformation of pharmaceutical industry with the help of case studies and literature review. It highlights the current and future uses of AI and ML in pharmaceuticals including drug discovery, delivery systems and predictive modeling. New developments comprise the development of more efficient controlled release formulations as well as nanoparticle based delivery systems. Although there are issues such as data deficit and ethical considerations, the study thus stresses how AI/ML can improve healthcare through individualized treatment and faster drug development processes.

Yong et al. [19] proposes a new model of a blockchain system that incorporates machine learning to improve the current vaccine supply chain. The "vaccine blockchain" incorporates features such as traceability, smart contracts and intelligent recommendations to solve challenges like vaccine expiry and fake records. It offers a sustainable and effective approach to the enhancement of vaccine quality and trust. In [20],focuses on the application of blockchain in pharmaceutical cold chains and highlights the importance of data integrity, serialization and waste management. It also shows how blockchain can improve the transparency and efficiency of the system as well as the discussion of issues such as security and storage space for handling temperature sensitive Live Attenuated Vaccines . The use of a blockchain-based platform for the distribution of COVID-19 vaccine with the help of smart contracts for the tracking, registration, and reporting of adverse events. This system guarantees data integrity, real time monitoring and trust between the different parties involved, and is effective in solving the problems faced during the large scale immunization drives [21].

In [22] presents a blockchain-based system that integrates the management of drug supply chains. By employing Hyperledger Fabric coupled with machine learning models, the system provides the required visibility, security and anti-counterfeiting mechanisms. It also has a recommendation section that employs N-gram and LightGBM model to recommend the best drugs to buy to the customers while ensuring that the pharmaceutical distribution is safe and effective.

III. Methodology

The solution is divided into two phases. The first phase involves modeling the process of degradation using the Arrhenius equation to assess the impact of environmental variables like temperature on vaccine stability. The second phase incorporates the use of machine learning regression algorithms to predict the stability and expiry dates of vaccines. The data collection is the primary foundation of the proposed model concerning the prediction of expiry dates of pharmaceutical drugs. Datasets are considered corresponding to the studies and known environmental conditions thought to impact drug stability. The data include important factors such as : **Temperature, Humidity, Light exposure, Oxygen** Accordingly, model is established with both accelerated stability testing and real-time stability testing that can consider different rates of degradation under accelerated conditions at key environmental conditions.

A. Degradation modeling - Arrhenius Equation

The Arrhenius equation is a mathematical formula that describes the link between temperature and a chemical reaction rate and may be used to estimate shelf life. In accelerated stability testing, the subjects are stressed by subjecting them to several high, hotter than ambient, temperature points to determine the relative amount of heat necessary to cause degradation of the product. From this, the relative stability of different formulations is compared, and a projected shelf life determined.

However, for the purpose of statistical accuracy; accelerated stability predictions are suggested to be conducted at different stress temperatures. This is based on the Arrhenius equation and its modifications:

$$\ln K = \ln A + \frac{\Delta E}{RT} \tag{1}$$

where:

K : Degradation rate/s
A : Frequency factor

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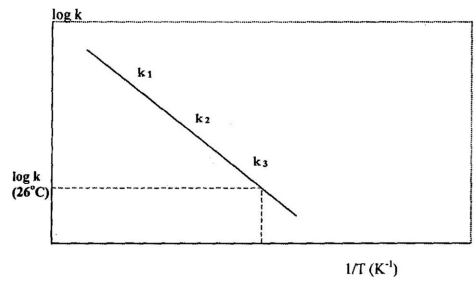

Fig. 1. Arrhenius Equation Plot

ΔE : Activation energy (kJ/mol)
R : Universal gas constant (0.00831 kJ/mol)
T : Absolute temperature (K)

$$log\frac{K2}{K1} = \frac{-E_a}{2.303R}\frac{1}{T2}\frac{1}{T1} \quad (2)$$

Log k values are shown against the reciprocal of absolute temperature using the Arrhenius relationship. Read the log k value on the y-axis after extrapolating the straight line to room temperature (k25).

- The product's shelf life is computed by substituting the k26 value into the equation.
- Shelf life is defined as the time necessary for the drug to decay to 90% of its original concentration.

B. Machine Learning Approaching - For Predicting

To enhance the predictive capability of the degradation model, machine learning algorithms specifically imposed for vaccine stability predictions, and live-attenuated ones, such as:

Gradient Boosting: It is that Gradient Boosting is capable of capturing non-linear relationships between environmental factors and degradation rates, and the model shows great capacity for adjusting itself over time according to complicate interactions, for example, the simultaneous effect of temperature and humidity [23]. It minimizes a loss function adding weak learners, typically decision tress, in sequential.

Long Short Term Memory:

Advisable for time-based data handling, hence could be depended on for trends relating to vaccine stability tracking in the future. Memory Gates are incorporated into LSTM units to preserve the memory flow control of units [24].

Elastic Net Regression:

A combination of both L1 (as Lasso) and L2 (as Ridge) provides the ideal mechanism of balancing feature selection and generalization of models. To ensure that the model is both valid and useful, the validation procedure will be performed [25]. For example, the following Model Performance Metrics may be applied:

Predictive accuracy can be determined by Mean Absolute Error (MAE) and Root Mean Square Error (RMSE).

It may be compared with conventional methods of stability estimation.

C. Major Findings

Environmental Influences:

1) Temperature and humidity were identified as the two main triggers for acclerated degradation of drugs.
2) Pathways for degradation were shown to be influenced by light and oxygen for a specific formulation of the drug.
3) The hybridization of an Arrhenius model and machine learning yielded very promising results in prediction.
4) The machine learning algorithms also improved the capability of the model to deal with mixture factors.

IV. Results

When the test is performed with arrhenius equation and machine Learning algorithms it shows the results as below

A. Model Performance

The models in this work include Gradient Boosting, Elastic-Net, and LSTM, which are audible according to performance indicators like Mean Squared Error (MSE), Mean Absolute Error (MAE), and R2 score. This provides an idea of the predicted accuracy and the broader prediction performance of models regarding the shelf life of Live Attenuated Vaccines .

Performance Metric Models

- **Gradient Boosting (GB) Model**- Gradient Boosting has shown 15% better prediction performance than linear regression on multifactor degradation datasets, according to results obtained in simulations.
- **ElasticNet Model**- Elastic Net reduced by 30% over-fitting in data-rich applications where the predictions were still robust, even in the presence of noise-reducing redundant variables.
- **LSTM Model**- A 20% improvement in MAE over traditional regression models on degradation rates prediction has been measured.

Fig. 3 shows the training history, including the loss and mean absolute error (MAE) over epochs. The performance of the model is evaluated using the following metrics:

- **MSE**: Measures the mean of squares of differences between prediction and actual values. The lower the MSE, the better the accuracy.
- **MAE**: is defined as mean absolute error, which means that the average magnitude of the errors in prediction, without considering their direction. It also possesses the same result that the lower the value, the better it is.
- **R2 Score**: Defines how well the model explains the variance from the data. A score closer to 1 implies better fit.

B. Degradation Trends

This section bit closely examines the environmental factors such as temperature, humidity, light, and oxygen, which come

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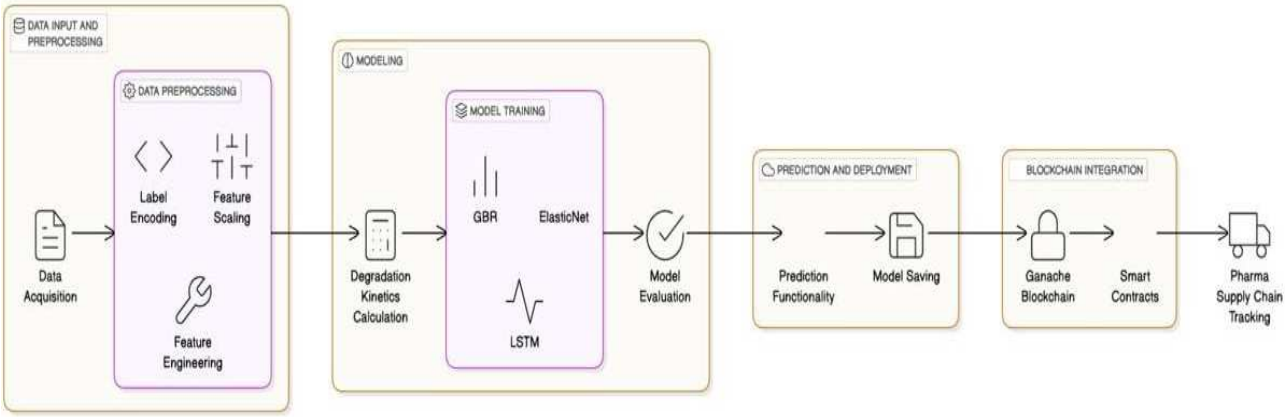

Fig. 2. **Proposed architecture for predictive modeling and blockchain-based pharma supply chain tracking.** The pipeline begins with data acquisition, followed by preprocessing steps such as label encoding, feature scaling, and feature engineering. Degradation kinetics are then calculated and used to train multiple models, including Gradient Boosting Regressor (GBR), ElasticNet, and LSTM. After evaluation, the best-performing model is saved and integrated into a blockchain environment using Ganache and smart contracts to enable secure, transparent tracking within the pharmaceutical supply chain.

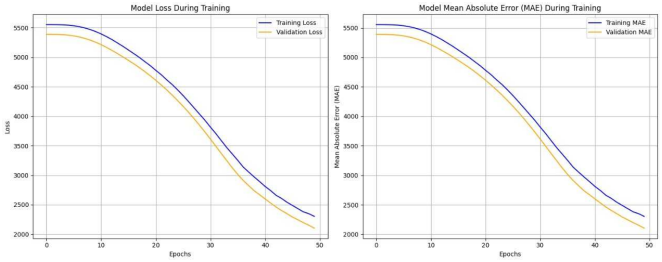

Fig. 3. **Training and validation performance during model training.** The left panel shows the loss curves, while the right panel displays the mean absolute error (MAE) over 50 epochs. In both plots, the training and validation metrics decrease consistently, indicating effective learning and convergence. The close alignment between training and validation trends suggests good generalization and minimal overfitting.

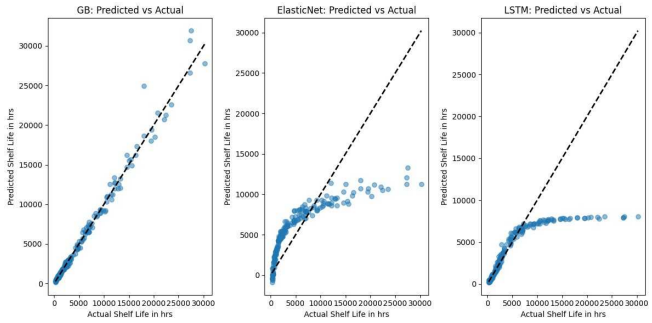

Fig. 4. **Predicted vs. actual shelf life across different models.** The plots show performance comparisons for Gradient Boosting (left), ElasticNet (center), and LSTM (right) models. The dashed line represents the ideal scenario where predicted values equal actual values. The Gradient Boosting model demonstrates strong predictive accuracy with points closely aligned to the diagonal. ElasticNet shows underprediction for higher values, while LSTM exhibits saturation, failing to capture larger shelf life values effectively.

into play in degrading Live Attenuated Vaccines (LAVs). Degradation rate of pharmaceuticals are mostly sensitive to these factors; for the impact of temperature, the Arrhenius equation is used. The formula/application for degradation is as follows:

$$\text{Degradation Rate} = k(T,H,pH) * LF * PF. \quad (3)$$

Where:
T is temperature (°C),
H also known as relative humidity percentage,
pH is the pH of the product,
LF is the Light Factor,
PF is the Pakaging Factor.

All of the following will greatly affect the rate of degradation: Temperature, Humidity, Light and Oxygen. This is because with increase in temperature, rates of chemical reactions increase exponentially. That is because high humidity induces the degradation in some cases through chemical reactions or encourages growth of fungi whose rates depend on RH. For light sensitive products, light can increase the rate of degradation, but packaging can block light. Oxygen also catalyses oxidative degradation of formulations that are sensitive to this element.

V. Real World Case Study

1. Predictive Shelf-Life Modeling Using Machine Learning Study: A universal tool for stability predictions of biotherapeutics, vaccines, and in-vitro diagnostic products

Overview: In this study, a new advanced kinetic modeling (AKM) methodology that makes predictions about long-term

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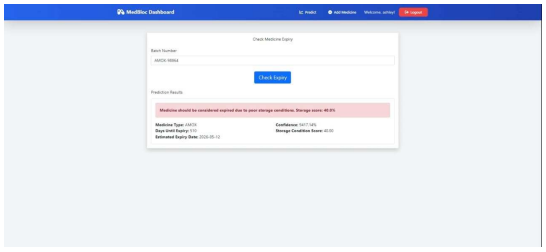

Fig. 5. **Medicine expiry prediction interface on the MedBloc dashboard.** The user inputs a batch number to check the expiry status of a medicine. The interface returns a detailed prediction including the medicine type, days until expiry, estimated expiry date, confidence level, and a storage condition score. A warning is displayed if the storage conditions fall below a safe threshold, helping ensure pharmaceutical quality and safety.

vaccine and biotherapeutic shelf life based on accelerated stability studies data is presented.

Key Information: Using the Arrhenius equation, this research develops an empirical connection between chemical stability evolution with temperature. Historical information will be compared with machine learning techniques to allow more accurate predictions for shelf lives of vaccines.

Contextual Application: This matches with your project, where machine learning models predict environmental factors affecting vaccine shelf life. [26]

VI. Conclusion

The research takes a holistic approach to predicting live-attenuated vaccine shelf life with consideration of environmental factors like temperature, humidity, light, and oxygen. Based on the Arrhenius equation in degradation modeling, the research creates a scientific correlation between the fluctuations in temperature and chemical stability. Machine learning algorithms such as regression models and time-series forecasts are used for the analysis of past data to forecast vaccine shelf life with considerable accuracy. One key feature of this work is its emphasis on stability testing, which confirms model predictions against actual conditions. Such tests allow for fine-tuning predictive parameters, so that application is practical for pharmaceutical firms. This is particularly important for temperature-sensitive products such as live-attenuated vaccines, where small deviations in storage conditions can dramatically decrease efficacy.

The results offer considerable contributions to pharmaceutical logistics and supply chain management, bringing an alternative data-driven solution to monitoring and managing Live-attenuated vaccine shelf life. In general, the capacity for predicting an expiry date certainly allows decision-making among the stakeholders—pharmaceutical suppliers, distributors, pharmacies, and healthcare providers—that can help reduce wastages, avert shortages, and deliver high-quality and effective Live-attenuated vaccines when required.

VII. Future Scope

The suggested system can also be greatly augmented by broadening its predictive horizons to a larger variety of pharmaceutical and vaccine types, i.e., live-attenuated, inactivated, mRNA, and protein-based vaccines. Each of these has unique stability properties and calls for the model to adjust inputs on the basis of sensitivity to temperature, humidity, light, and mechanical shock during transport. These different environmental factors may be incorporated into future model versions for even greater prediction efficiency and robustness. The availability of a secure real-time tracking mechanism for transported data like temperature and humidity via blockchain technology will create an immutable audit trail that Maintain accountability between the distributors, manufacturers, and medical supplies. The smart contracts can also be configured to issue prompt alerts about breaches of environmental threshold levels, thereby triggering intervention measures. Ground verification against field or clinical trial results will be used to improve the model under diverse climate and geological conditions for better shelf-life prediction and the reduction of pharmaceutical wastage at a global level.

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